

Abel Mendoza

Robotics & Embedded Systems Engineer

Anaheim, CA

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Professional Summary

Robotics and embedded systems engineer with hands-on experience deploying, repairing, validating, and integrating autonomous robotic and industrial hardware systems in real-world environments. Skilled in ROS2, embedded Linux, hardware validation, sensor fusion, industrial communications, and full-stack system integration across software, electronics, networking, and control layers. Proven ability to diagnose failures and restore systems under live operational pressure. Targeting aerospace, defense, robotics, and embedded autonomy roles. U.S. citizen eligible for ITAR/EAR work.

Technical Skills

Autonomy & Robotics: ROS2 Humble, PX4 Autopilot (SITL), MAVLink, MAVSDK, SLAM, Kalman filtering, obstacle avoidance

Perception: OpenCV, ArUco tracking, Intel RealSense D455, YOLOv8 (Jetson TensorRT), depth sensing

Embedded & Electronics: Raspberry Pi 4B, NVIDIA Jetson Orin Nano, STM32, PIC32, FPGA tooling, UART, SPI, I2C, PWM, GPIO

Electrical Test & Validation: Multimeters, optical power meters, programmable electronic loads, DC-DC converter testing, relay logic validation, continuity testing

Systems & Networking: Linux (Ubuntu 22.04), Docker, Tailscale VPN, WebSockets, TCP/UDP, distributed systems

Programming: Python, C++, Go, TypeScript, JavaScript, Bash

Tools: Git, VS Code, FastAPI, Next.js, React, CycloneDDS, MPLAB ICD3, Quartus

Professional Experience

Electronics Test & Validation Technician

Apr 2026 – Present

RLH Industries, Inc. — Orange, CA

- Performed validation, calibration, and troubleshooting of fiber optic communication modules, verifying electrical → optical → electrical signal integrity across telecom and industrial communication systems
- Measured voltage, current, resistance, and continuity using multimeters, programmable electronic loads, and bench power supplies
- Validated optical transmission systems at 850nm and 1310nm wavelengths using calibrated optical power meters and dBm measurements
- Executed DC-DC power converter testing and calibration including voltage regulation, load testing (~95–100W), tolerance verification ($\pm 0.5\%$), and thermal drift observation
- Calibrated relay threshold detection and alarm logic circuits (VR1/VR2), validating LED indicators, switching behavior, and fault conditions
- Performed firmware flashing, embedded hardware bring-up, and web interface deployment for industrial Ethernet relay and I/O systems using MPLAB ICD3, PIC32 microcontrollers, MPFS2, USB Blaster, and Quartus
- Conducted board-level troubleshooting and failure isolation using continuity testing, embedded network diagnostics, and known-good reference hardware (“golden boards”)
- Assisted with FPGA programming, board-level validation, and integration testing for industrial communication and supervisory control systems

Robotics Deployment Technician

Jan 2025 – Dec 2025

Armstrong Robotics — Hollywood, CA

- Maintained 24/7 deployment of an autonomous robotic system in a high-volume production environment as the sole on-site technical owner
- Assembled and integrated new hardware components to improve system autonomy and operational reliability

- Performed repairs across mechanical, electrical, and control systems during live robotic operation
- Diagnosed and resolved failures across hardware, perception, and control systems under real-time constraints
- Replaced and calibrated Robotiq grippers on UFactory robotic arms and validated performance under production load
- Calibrated vision systems using ArUco markers and supported live iteration with engineering teams
- Acted as liaison between operators, staff, and engineers during system failures and recovery

Field Robotics Agent

Mar 2025 – May 2025

Serve Robotics — Los Angeles, CA

- Supported deployment and expansion of autonomous delivery robots across multiple LA regions
- Diagnosed navigation, sensor, and connectivity issues during real-world operation and validated fixes under time pressure
- Collaborated with engineering teams to surface edge cases and improve system reliability

Technical Projects

Omega-1 — Distributed Autonomous Robotics Platform

Raspberry Pi 4B | Jetson Orin Nano | ROS2 | FastAPI | Next.js

- Architected a distributed robotics system across multiple machines enabling real-time control, perception, and monitoring over a VPN-secured network
- Implemented GPS-free localization using Kalman filter sensor fusion combining odometry with ArUco-based landmark corrections
- Built real-time obstacle avoidance system with ultrasonic sensing and dynamic motion-state recovery
- Designed low-latency video streaming pipeline using custom MJPEG proxy infrastructure
- Integrated YOLOv8 object detection on Jetson Orin with optimized compressed streaming
- Debugged PWM clock interference corrupting servo signals and resolved through timing validation and signal clamping
- Built full-stack control dashboard with React and Xbox controller teleoperation support

PX4 Autonomous Mission Stack

ROS2 | PX4 | Gazebo

- Built UAV autonomy pipeline with offboard control, navigation logic, and mission-state handling
- Implemented obstacle avoidance and telemetry visualization using ROS2 and MAVSDK

Jetson RealSense Perception Stack

Jetson Orin Nano | ROS2

- Built RGB-D perception pipeline with depth streaming, IMU integration, and obstacle detection
- Developed diagnostics to validate sensor data, frame rates, and runtime system performance

Education

California State University, Fullerton

May 2024

Bachelor of Science — Computer Science

Rio Hondo Community College

Electrical and Electronics Coursework

Additional

FAA Part 107 Certified

U.S. Citizen — ITAR/EAR Eligible

Comfortable with field operations, hardware deployment, and physically demanding environments

Languages: English (native), Spanish (conversational)